

# Early Detection of Alzheimer's Disease Using Machine Learning on Clinical and Lifestyle Data

Poorvika Bhatia

Dept. of Electronics and Communication Engineering  
Bharati Vidyapeeth (Deemed to be University), College of Engineering  
Pune, Maharashtra  
[purvibhatia3732@gmail.com](mailto:purvibhatia3732@gmail.com)

**Abstract-** Alzheimer's disease is a progressive brain disorder that impairs memory, thinking, and organizational skills, and is the leading cause of dementia worldwide. It affects nearly 24 million people, with prevalence increasing significantly after age 65. Common symptoms include memory loss, confusion, difficulty with decision-making, language impairment, and personality changes, all of which severely impact daily life. While no cure exists, early detection and timely treatment can slow progression and improve quality of life. This research proposes the use of Machine Learning (ML) for early and accurate detection of Alzheimer's disease using clinical and lifestyle practical for everyday clinical use. The expected outcome is a lightweight, accessible model that can support healthcare providers in early diagnosis and treatment decisions. This research highlights the potential of ML in developing cost-effective and scalable solutions for detecting Alzheimer's and related brain disorders.

## I. INTRODUCTION

Alzheimer's disease is a brain disorder that slowly destroys memory and thinking skills, and eventually, the ability to carry out the simplest tasks. In most people with Alzheimer's, symptoms first appear later in life. Estimates vary, but experts suggest that more than 6 million Americans, most of them age 65 or older, may have Alzheimer's. Alzheimer's is currently ranked as the seventh leading cause of death in the United States and is the most common cause of dementia among older adults. Dementia is the loss of cognitive functioning like thinking, remembering, reasoning and behavioral abilities to such an extent that it interferes with a person's daily life and activities. Dementia ranges in severity from the mildest stage, when it is just

beginning to affect a person's functioning, to the most severe stage, when the person must depend completely on others for help with basic activities of daily living. Scientists continue to unravel the complex brain changes involved in Alzheimer's. Changes in the brain may begin a decade or more before symptoms appear. During this very early stage of Alzheimer's, toxic changes are taking place in the brain, including abnormal buildups of proteins that form amyloid plaques and tau tangles. Previously healthy neurons stop functioning, lose connections with other neurons, and die. Many other complex brain changes are thought to play a role in Alzheimer's as well.

Traditional diagnostic methods for Alzheimer's Detection often rely on clinical assessments, neuropsychological tests, and advanced brain imaging techniques such as MRI and PET scans. These methods are not only expensive and time-consuming but also inaccessible in low-resource clinical settings. Furthermore, early symptoms like forgetfulness or personality changes are often overlooked which leads to delayed diagnosis. But Machine Learning (ML) provides an alternative by enabling early and accurate detection of Alzheimer's disease using easily available patient health and lifestyle data. By analyzing features such as body mass index (BMI), diet quality, history of head injuries, hypertension, and behavioral changes, ML models can identify hidden patterns that may indicate the onset of Alzheimer's.

In this research, we propose a lightweight and easy to use ML model for Alzheimer's Detection. Unlike Deep learning which require complex imaging data, our method is based on simple algorithms like Random Forest which is faster and suitable for everyday clinical use. Our approach

consists of a) Collection and preprocessing of publicly available Alzheimer's dataset based on lifestyle and clinical features; b) Implementation of feature selection to remove unnecessary variables; c) Splitting the dataset for training and testing and applying Random Forest Algorithm; d) Evaluation of model using accuracy, confusion matrix, and classification report. This paper is organized as follows: Section II presents related work, Section III describes the methodology, Section IV discusses the experimental results, Section V concludes the study and outlines future directions.

## II. RELATED WORK

Machine Learning has become one of the most popular approach for early detection and diagnosis of Alzheimer's Disease. Traditional diagnostic methods are based on medical imaging such as Magnetic Resonance Imaging (MRI) and Positron Emission Tomography (PET) scans but these methods are expensive and less accessible. To overcome these limitations, researchers started focusing on machine learning techniques to build predictive models that can utilize both tabular (non-imaging) and imaging data.

Kim et al. [1] created a Random Forest model that used MRI scans of the brain to classify patients with Alzheimer's disease. Their study showed that Random Forest, as an ensemble learning method, could give very good accuracy for this task. In a similar way, Lebedev et al. [2] also tested Random Forest on MRI-based features and found better results when they combined information from different types of scans. Still, both methods depended heavily on costly brain imaging, which makes them harder to use in everyday healthcare. Gupta and Kumar [3] also suggested a modified Random Forest (m-RF) with improved feature extraction and reported an accuracy of 96.4%. While these methods gave strong results, they also became more complicated and needed more computation, which limits their practical use. Islam and Zhang [4] discussed deep learning models and observed that using both imaging and non-imaging data together can give the best accuracy, although these models are often hard to interpret.

All these approaches show that Random Forest and other machine learning techniques can give high accuracy in detecting Alzheimer's. And it depends on either imaging techniques or complex optimizations. But our approach uses simple clinical and lifestyle data and this makes our approach more practical, cost-effective, and useful for early

detection, especially in healthcare systems with limited resources.

## III. METHODOLOGY

This research focuses on building a machine learning model to detect Alzheimer's disease using patient health and lifestyle data instead of costly imaging techniques. The complete process consists of five main steps: dataset collection, preprocessing, model training, evaluation, saving the model and creating a web app.

### A. Dataset Collection

The dataset used in this study was obtained from Kaggle, a publicly available online repository for datasets. It consists of features like Age, Gender, BMI, diet quality, sleep quality, Smoking, Alcohol Consumption, Diabetes, Blood Pressure, Cholesterol, Family History Alzheimer's, Head Injury, Hypertension, Forgetfulness, Personality Changes, Diagnosis, etc. The target variable indicates whether a patient has been diagnosed with Alzheimer's disease. The dataset consists of 2149 records and 35 features, providing sufficient data for training and testing the machine learning models.

### B. Data Preprocessing

Before training the model, data preprocessing was applied to check whether the dataset contains missing values or not and to drop unnecessary features. This step is important because it improves the accuracy of model. After preprocessing, the dataset consists of 2149 records and 17 features.

### C. Model Training

For this study, the main algorithm used was the Random Forest classifier, chosen because it can handle high-dimensional data, reduce overfitting, does not require encoding, and give interpretable results. Random Forest Algorithm is a tree based algorithm which consists of multiple decision trees. Using Scikit learn library (from `sklearn.model_selection import train_test_split`, from `sklearn.ensemble import RandomForestClassifier`, from `sklearn.metrics import classification_report, confusion_matrix, accuracy_score`), the dataset was split into 80% training and 20% testing for evaluation.

### D. Model Evaluation

The model was evaluated using several metrics like accuracy score (the percentage of correct

predictions), confusion matrix (a table showing correct and incorrect predictions for each class), classification report (A summary showing how well the model predicts each class, including precision, recall, and F1-score). These metrics were used to check how well the model worked.

#### E. Saving the model and creating a web app

After training and evaluating the Random Forest model, the model was **saved** using Python's Pickle library, allowing it to be reused without retraining. A web application was then created

using Streamlit, which provides an easy-to-use interface for users to input patient data and get predictions in real time. This step makes the model practical for real-world use by allowing doctors or healthcare workers to interact with it directly.

## Using Random Forest Classifier

```
[16]: from sklearn.ensemble import RandomForestClassifier

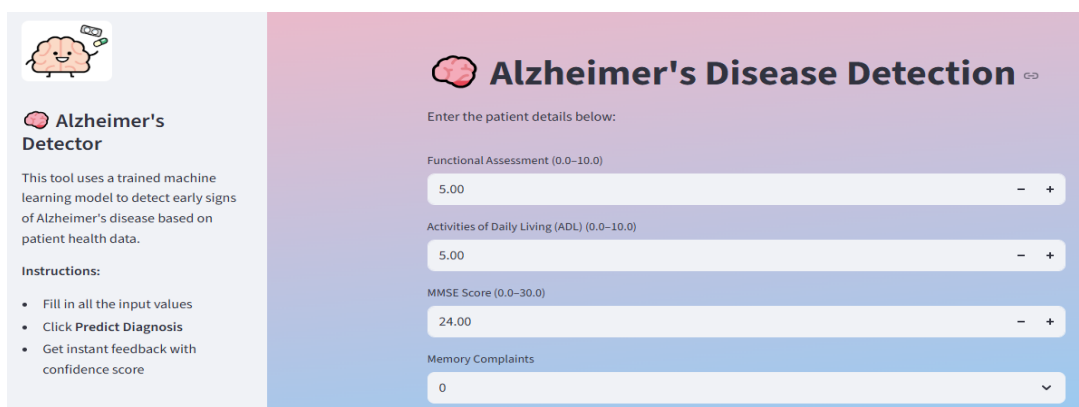
[17]: model = RandomForestClassifier(n_estimators=150,criterion='gini')

[18]: model.fit(x_train,y_train)

[18]: ▼ RandomForestClassifier ⓘ ?
      RandomForestClassifier(n_estimators=150)

[19]: y_pred = model.predict(x_test)
```

Fig. 1. Model Training using Random Forest Classifier



The screenshot shows a web application titled "Alzheimer's Disease Detector" on the left and "Alzheimer's Disease Detection" on the right. The left panel includes a brain icon, the title, a description of the tool, and instructions. The right panel is a form for entering patient details, including Functional Assessment, Activities of Daily Living (ADL), MMSE Score, and Memory Complaints, each with a text input field and a submit button.

**Alzheimer's Disease Detector**

This tool uses a trained machine learning model to detect early signs of Alzheimer's disease based on patient health data.

**Instructions:**

- Fill in all the input values
- Click **Predict Diagnosis**
- Get instant feedback with confidence score

**Alzheimer's Disease Detection**

Enter the patient details below:

Functional Assessment (0.0–10.0)  
5.00

Activities of Daily Living (ADL) (0.0–10.0)  
5.00

MMSE Score (0.0–30.0)  
24.00

Memory Complaints  
0

Fig. 2. A web app for Alzheimer's Disease Detection using streamlit

#### IV. EXPERIMENTAL RESULTS

The proposed Random Forest model was evaluated on the Kaggle Alzheimer's dataset using multiple metrics. The model achieved an accuracy of 94.6%, demonstrating strong performance in detecting Alzheimer's disease.

##### A. Confusion Matrix

Fig. 3 shows the confusion matrix for the Random Forest model. The matrix clearly indicates that the model correctly identified the majority of Alzheimer's positive and negative cases, with only a few misclassifications.

```
Confusion Matrix:
[[272  5]
 [ 18 135]]
```

Fig. 3. Confusion Matrix for Random Forest Classifier

##### B. Classification Report

Fig. 4 shows the classification report for the Random Forest model. It shows the values of precision, recall, F-1 score and support of the model.

```
Classification Report:
              precision    recall  f1-score   support

0               0.94        0.98        0.96         277
1               0.96        0.88        0.92         153

accuracy               0.95
macro avg              0.95        0.93        0.94
weighted avg           0.95        0.95        0.95
```

Fig. 4. Classification Report for Random Forest Classifier

##### C. Streamlit Web Application

A web application was developed using Streamlit. This application allows user to input patient information such as MSME score, Activities of Daily Living (ADL), Functional Assessment, Memory Complaints and on the basis of these inputs our model will predict whether the patient has Alzheimer's or not. Fig. 2 shows the interface of streamlit web application.

#### V. CONCLUSION AND FUTURE WORK

This study developed a Random Forest-based machine learning model for the early detection of Alzheimer's disease using clinical and lifestyle data. The model achieved 94.6% accuracy and can be accessed through a Streamlit web application, making it easy for healthcare professionals to use. The approach is cost-effective, does not require expensive imaging, and provides quick, reliable predictions.

Future work can include: a) Expanding the dataset with more patient records; b) Testing other ML algorithms to compare with existing model and improve the accuracy; c) Enhancing the web app with more user friendly features like integrating an AI powered assistant or chatbot to guide users and provide health related recommendations alongside predictions; d) Also integrating it with hospital databases. Overall, this research demonstrates that machine learning can make Alzheimer's detection more accessible, faster, and practical for clinical use.

#### REFERENCES

- [1] J. Kim, et al., "Random Forest-Based Classification of Alzheimer's Disease using Structural MRI Features," *Frontiers in Aging Neuroscience*, vol. 12, pp. 1–9, 2020.
- [2] A. Lebedev, et al., "Random Forest Ensembles for Improved Classification of Alzheimer's Disease Based on MRI Features," *NeuroImage: Clinical*, vol. 22, pp. 101–113, 2019.
- [3] R. Gupta and S. Kumar, "An Optimized Random Forest Model for Early Detection of Alzheimer's Disease," *International Journal of Computer Applications*, vol. 176, no. 3, pp. 12–18, 2021.
- [4] M. Islam and Y. Zhang, "Deep Learning for Alzheimer's Disease: A Comprehensive Survey," *IEEE Reviews in Biomedical Engineering*, vol. 14, pp. 263–276, 2021.